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Signature of the Faculty

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```
From pyspark.sql import SparkSession
Spark = SparkSession.builder.master("local[*]").appName
("RDD Example").getOrCreate()
Sc = Spark.SparkContext
RDD = SD.parallelize([10,20,30,40,50,60,70,80])
print(RDD.collect())
print("partition:", RDD.getNumPartitions())
Spark.Stop()
```

Q.1) Perform a practical to create RDD using parallelize() function in apache Spark.

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Q:2 Practical : 2

Perform a practical to read text datasets into RDD Using Apache Spark.

```
! pip install pyspark

from pyspark import SparkContext
sc = SparkContext.getOrCreate()

data = [
    "101, prateek, 85",
    "102, Ali , 78",
    "103, ASif , 92",
    "104, Raheem , 75"
    "105, ROx , 88",
]

rdd = sc.parallelize(data)
print("content of RDD:")
for row in rdd.collect():
    print(row)

print("\n Total Rows:", rdd.count())

CSV_rdd = rdd.map(lambda row: Split(","))

print("\n RDD after Splitting :")
for row in CSV_rdd.collect():
    print(row)
```

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Q.3 Perform practical to read CSV datasets into RDDs Using Apache Spark.

```
import pandas as pd
from pyspark import SparkContent

! Pip install pyspark

n = int (input ("Enter the number of Student :"))
data = []

for i in range (n):
    print (f"\n Enter details for Student {i+1} ")

    Student_id = input ("Enter ID:")
    name = input ("Enter NAME:")
    marks = input ("Enter marks:")

    data.append ([Student_id, name, marks])

CSV_file = "Student_data.csv"
df = pd.DataFrame (data, clowmns = ["ID", "Name", "marks"])
df.to_csv (CSV_file, index = false)

print ("\n CSV file created Successfully !")
print (df)

try :
```

```
    sc = SparkContext.getOrCreate()
except exception as e:
    print(f"could not get or create SparkContext: {e}")

rdd = sc.textFile(csv_file)

csv_rdd = rdd.map(lambda row: row.split(","))

print("\n Content of CSV RDD:")

for row in csv_rdd.collect():
    print(row)

print("\n Total Rows (including headers:", csv_rdd.
count())
```

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Q.4 Perform a practical to implement RDD transformations Using repartition Operation.

Q:5 perform a practical to implement RDD transformations Using coalesce Operation.

```
from pyspark.sql import SparkSession

spark = SparkSession.builder\
    .appName("RDD Repartition practical")\
    .master("local[*]")\
    .getOrCreate()

sc = spark.sparkContext

data = list(range(1, 10))
rdd = sc.parallelize(data, 2)

print("Original RDD:")
print(rdd.collect())

print("\n Original Number of partitions:")
print(rdd.getNumPartitions())

print("\n Original Number of partitions:")
print(rdd.getNumPartition())

new_rdd = rdd.coalesce(4, shuffle = True)
```

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```
print("\n Number of partition after Repartition:")
print(new_rdd.getNumPartitions())

print("\n RDD After Repartition:")
print(new_rdd.collect())

print("\n Data in each partition:")
partition_data = new_rdd.glom().collect()

for i, partition in enumerate(partition_data):
    print("partition", i+1, ":", partition)

Spark.stop()
```

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